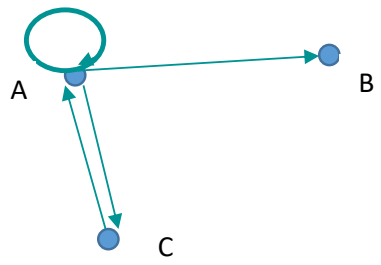
Lesson 5 - Shortest Path Problem

5.1 Directed graph or digraph

Question 56

Level of difficulty: E

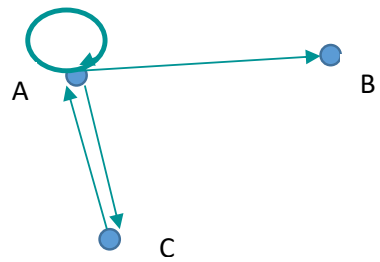
Given the digraph, enter the input and output degrees for all nodes.



Question 57

Level of difficulty: E

Given the digraph, draw its shadow.



Solution question 56

Solution hint: Look up how many edges (arrows) lead in and out of the respective nodes.

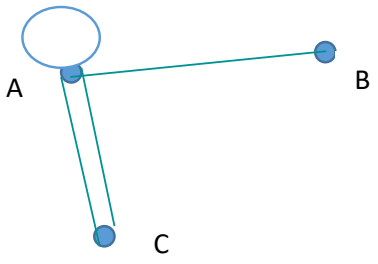
Solution:

	Input degree	Output degree
A	2	3
B	1	0
C	1	1

Solution question 57

Solution hint: The shadow of a digraph is an undirected graph.

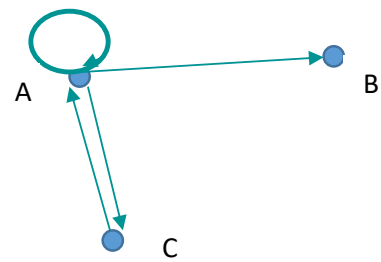
Solution:



Question 58

Level of difficulty: M

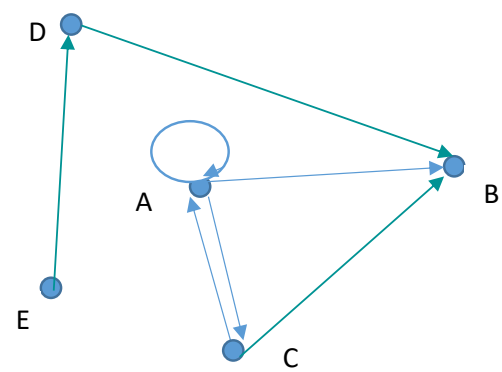
Specify the adjacency matrix of the given digraph.



Question 59

Level of difficulty: M

Specify the adjacency matrix of the given digraph.



Solution question 58

Solution hint: The adjacency matrix specifies how many directed edges from one node enter each of the other nodes.

Solution: A = $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$

Solution question 59

Solution hint: The adjacency matrix specifies how many directed edges from one node enter each of the other nodes.

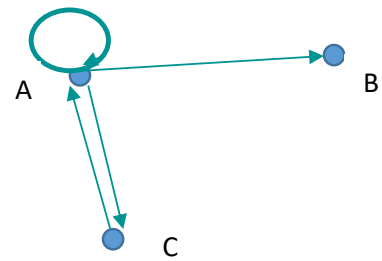
Solution: A = $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$

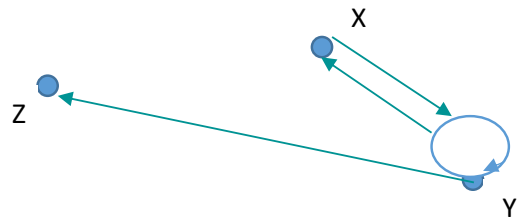
Question 60

Level of difficulty: D

Are the two graphs given isomorphic to each other?

If yes, state the bijective mapping that both graphs transfer into each other.





Solution question 60

Solution hint: Give a bijective mapping of the set of nodes of the first graph to the set of nodes of the second graph.

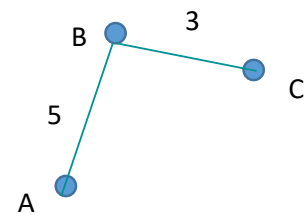
Solution: The two graphs are isomorphic. A -> Y, B -> Z, C -> X

5.2 Weighted graph

Question 61

Level of difficulty: E

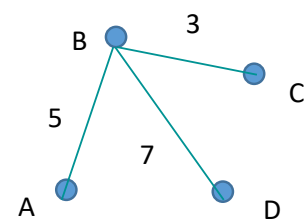
For the following graph, give the weight matrix C.



Question 62

Level of difficulty: E

For the following graph, give the weight matrix C.



Solution question 61

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & 5 & \infty \\ 5 & 0 & 3 \\ \infty & 3 & 0 \end{pmatrix}$

Solution question 62

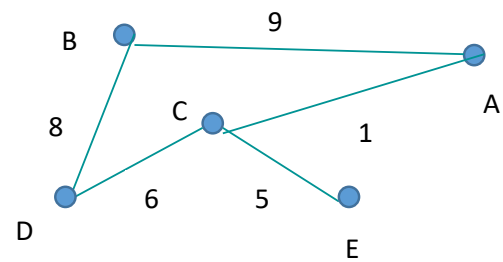
Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & 5 & \infty & \infty \\ 5 & 0 & 3 & 7 \\ \infty & 3 & 0 & \infty \\ \infty & 7 & \infty & 0 \end{pmatrix}$

Question 63

Level of difficulty: M

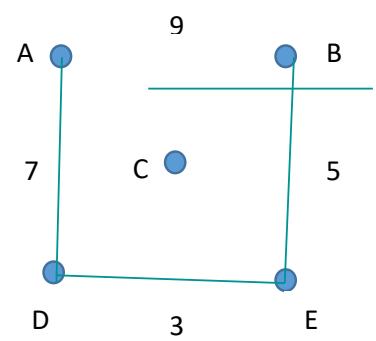
For the following graph, give the weight matrix C.



Question 64

Level of difficulty: M

For the following graph, give the weight matrix C.



Solution question 63

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: $C = \begin{pmatrix} 0 & 9 & 1 & \infty & \infty \\ 9 & 0 & \infty & 8 & \infty \\ 1 & \infty & 0 & 6 & 5 \\ \infty & 8 & 6 & 0 & \infty \\ \infty & \infty & 5 & \infty & 0 \end{pmatrix}$

Solution question 64

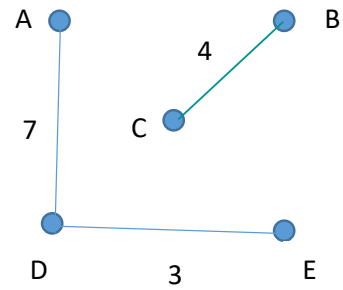
Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: $C = \begin{pmatrix} 0 & 9 & \infty & 7 & \infty \\ 9 & 0 & \infty & \infty & 5 \\ \infty & \infty & 0 & \infty & \infty \\ 7 & \infty & \infty & 0 & 3 \\ \infty & 5 & \infty & 3 & 0 \end{pmatrix}$

Question 65

Level of difficulty: S

For the following graph, give the weight matrix C.



Solution question 65

Solution hint: Sort the nodes in alphabetical order and use the sorting when building the matrix.

Solution: C = $\begin{pmatrix} 0 & \infty & \infty & 7 & \infty \\ \infty & 0 & 4 & \infty & \infty \\ \infty & 4 & 0 & \infty & \infty \\ 7 & \infty & \infty & 0 & 3 \\ \infty & \infty & \infty & 3 & 0 \end{pmatrix}$

5.3 Dijkstra's algorithm

Question 66

Level of difficulty: E

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 4 & 5 \\ \infty & 0 & 2 \\ 3 & 1 & 0 \end{pmatrix}$, where the edges are sorted alphabetically, determine the shortest paths from the starting node A to all other nodes.

Question 67

Level of difficulty: E

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 6 \\ 1 & 0 & \infty \\ \infty & 3 & 0 \end{pmatrix}$, where the edges are sorted alphabetically, determine the shortest paths from the starting node A to all other nodes.

Solution question 66

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	4
C	A -> C	5

Node	1	2
A		
B	**4** A	
C	5 A	**5** A
M	A, B	A, B, C
P	C	

Solution question 67

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	5
C	A -> C	6

Node	1	2
A		
B	**5** A	
C	6 A	**6** A
M	A, B	A, B, C
P	C	

Question 68

Level of difficulty: M

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 7 & 8 \\ 2 & 0 & 1 & 4 \\ 3 & \infty & 0 & 1 \\ 4 & 3 & \infty & 0 \end{pmatrix}$, where the edges are sorted alphabetically: A, B, C, D. Determine the shortest paths from the starting node A to all other nodes.

Question 69

Level of difficulty: M

Given a graph with its weight matrix C = $\begin{pmatrix} 0 & 5 & 3 & \infty \\ \infty & 0 & 2 & \infty \\ \infty & 1 & 0 & 7 \\ 4 & \infty & \infty & 0 \end{pmatrix}$, where the edges are sorted alphabetically: A, B, C, D. Determine the shortest paths from the starting node A to all other nodes.

Solution question 68

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> B	5
C	A -> B -> C	6
D	A -> B -> C -> D	7

Node	1	2	3
A			
B	**5** A		
C	7 A	**6** B	
D	8 A	8 A	**7** C
M	A, B	A, B, C	A, B, C, D
P	C, D	D	

Solution question 69

Solution hint: Use Dijkstra's algorithm.

Solution:

Node	Shortest path	Shortest distance
B	A -> C -> B	4
C	A -> C	3
D	A -> C -> D	10

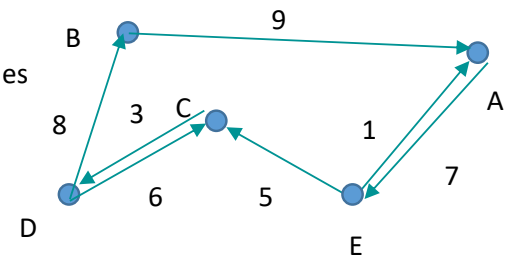
Node	1	2	3
A			
B	5 A	**4** C	
C	**3** A		
D	∞	10 C	**10** C
M	A, C	A, C, B	A, C, B, C
P	B	D	

Question 70

Level of difficulty: D

Given the following weighted digraph, arrange the nodes in alphabetical order.

Using the Dijkstra's algorithm determine the shortest paths from node A to all other nodes.



Solution question 70

Solution hint: Set up the weight matrix C and use Dijkstra's algorithm.

Solution: $C = \begin{pmatrix} 0 & \infty & \infty & \infty & 7 \\ 9 & 0 & \infty & \infty & \infty \\ \infty & \infty & 0 & 3 & \infty \\ \infty & 8 & 6 & 0 & \infty \\ 1 & \infty & 5 & \infty & 0 \end{pmatrix}$

Node	Shortest path	Shortest distance
B	A -> E -> C -> D -> B	23
C	A -> E -> C	12
D	A -> E -> C -> D	15
E	A -> E	7

	1	2	3	4
A				
B	∞			**23** D
C	∞	**12** E		
D	∞		**15** C	
E	**7** A			
M	A, E	A, E, C	A, E, C, D	A, E, C, D, B
P				